

Natural Vegetation and Wildlife

- India is rich in biodiversity and is among the 12 megabiodiversity countries in the world.
- 4700 species of plant, and ranks 10th in the world in terms of diversity.
- 15000 species → ~~Flaming~~ Flowering plant.
- 90,000 species present in India.
- Natural Vegetation: It refers to plant community which has grown naturally without any human disturbance and is left undisturbed.
- ⇒ Wildlife: All the wild animals living in their natural environment.
- ⇒ Species may be:
 - 1) Indigenous Species.
 - Also called as native species.

- That naturally occur in a particular region or area, e.g. Bengal Tiger, Neam.

2) Exotic Species: Those species which are introduced from another area.

e.g. Cantara Camora.

3) Endemic Species: That are present in a particular geographical location.

e.g. Hangul. (J&K).

→ Flora: Species of plants are collectively called as flora.

→ Fauna: Species of animals are collectively c/a Fauna.

⇒ Types of Vegetation:

1) Tropical Evergreen forest.

2) Tropical deciduous forest.

3) Tropical thorn forest or scrubs.

4) ~~Mon~~ Mountain forest.

5) Mangrove forest.

1) Tropical Evergreen Forest

⇒ Location: Present in the areas of Western Ghats, Islands of Andaman and Nicobar, Upper parts of Assam and T.N. Coast.

→ They flourish in area where rainfall is high. (more than 200 cm).

→ Characteristics: Height of trees is high (600 m or more).

→ They do not shed their leaves, thus forest remains green throughout the year.

→ It has all kinds of trees, shrubs, creepers, etc.

Thus multi-layered structure.

→ Commercially imp. Trees - Ebony, Mahogany,

Rubber, Rosewood, Cinchona.

→ Animals - Elephant, Monkey, Horned rhino (Assam W.B)

② Tropical Deciduous Forest

- Most Widespread forest.
- Also called as monsoon forest.
- Rainfall: 70-200 cm

- 2 Types -

1) Moist deciduous forest.

2) Dry deciduous Forest.

⇒ Moist Deciduous Forest	Dry Deciduous Forest
- <u>Rainfall</u> : 100-200 cm	- <u>Rainfall</u> - 70-100 cm
- <u>Location</u> - Eastern part of the country, N-Eastern states, along Himalayan foothill, Jarkhand, Odisha,	- <u>Location</u> - Rainer part of Peninsular Plateau, Bihar, UP, etc.

eastern slope of western
ghat.

- Plantation: Teak, sal, peepal,
Neem, etc.

- Plantation: Teak → dominant
specie.

Bamboo, sal, Mulberry,
kusum, khair →
Commercial plant.

→ Large part of these forest
are cleared for agriculture.

- Animals: Lion, tiger, deer, and elephant, variety
of birds, snakes, lizards, etc.

③ Thorn forest and scrubs

- Rainfall: Less than 70cm

- Vegetation consists of thorny trees and bushes,
to conserve water.

- Location: Present in N-Western part of India, in
the semi-arid area of Gujarat, Rajasthan, M.P.

Chettisgarh, V.P, Maryana, etc.

→ Plantation - Acacia, Palm tree, Euphorbia, and cactus.

- Scattered Vegetation, long deep roots to absorb water

- Stems are succulent (~~thick~~ thick and fleshy to store water).

- Leaves are thick to minimize water loss.

- Animals: rats, mice, rabbit, wolf, horses, camels, ~~the~~ lion, tiger, etc.

(4) ⇒ Mangrove Forest -

- Tidal forest, which are found in coastal areas influenced by tides.

- Location - They are found in coastal areas, ~~influenced~~

→ Location - They are found on the deltas of Ganga, Mahanadi, Kaveri, Krishna, Godavari.

→ Plantation - Mangrove trees are dominant here.

They develop breathing roots outside the soil surface.

→ Ganga Brahmaputra Deltas - Sundari tree is present here. It provides durable timber.

- Palm, Coconut, Agar, etc.

- Animals Royal Bengal Tiger, Gharial, Snakes, etc.

(5) → Montane Forests:

• Vegetation found in mountainous areas.

→ As the altitude increases, ~~temp~~ temp. decreases and vegetation changes.

→ On the basis of altitude, vegetation is divided into following types:

1) The wet temperate forests:

- Altitude: 1000 - 2000 m.

- Present ~~in~~ on low hills.

- Evergreen, broad leaved trees: e.g., Oak tree and chestnut.

2) Temperate Forests

- ~~Altitude~~ Altitude: 1500 - 3000 m.

- Location- Southern slope of Himalayas, high altitudes of S. India, N-Eastern part of India.

- Vegetation: Pine, Deodar, Cedar, Spruce, Silverfir.
(Coniferous Trees).

3) Alpine Forest

- Altitude: Above 3600 m.
 - It consists of Temperate vegetation (Pine, Silverfir) and grasslands.
 - As we move towards the snowline, shrubs, scrubs and grasses are present.
 - As we move above, lichens and mosses are ~~forest~~ present. (Tundra Vegetation).
 - Grasslands here ^{are} used by ~~Gujjar~~ Gujjars and bakormals for grazing.
- Animals: Kashmiri stag (Hangul), Spotted deer, Snow-leopard, Wild sheep, Tibetan Antelope, Bear, Sheep, Goat, etc.

⇒ Wildlife

- India is rich in fauna species.

90,000 species of animals
2000 species of birds
2546 species of Fishes

→ consists of 13% of the World's species.
12% of the World's species.

- 5-8% of the World's amphibians, reptiles and Mammals belongs to India.

⇒ Important Wildlife:-

1) Elephant - Majestic mammal.

- Found in wet forests of Assam, Karnataka and Kerala.

2) One-horned rhinoceros:-

- Endemic to India.

- Found in marshy and swampy areas of Assam and W. Bengal.

3) Camel - Present in Thar Desert.

4) Lion - Present in Gir forests of Gujarat.

5) Tiger - Present in M.P., Sunderbans of W. Bengal, Himalayan region.

- India is the only country with natural population of tiger and lion.

6) Leopard - Belongs to cat family.

⇒ Fauna in Himalayas:-

• These animals have adapted themselves to survive in cold temperature.

- Ladakh: e.g. Yak, shaggy ~~horned~~ horned wild ox (weighs around 1 ton).

Tibetan Antelope, the Bharal (blue sheep), wild sheep, the Kiang, Ibex, Snow Leopard, Rare Red Panda.

→ India: Rich variety of birds as well.
e.g. Peacock, Pheasants, Ducks, Pigeons, etc.

→ Water: Crocodile, Gharial, Turtle and different species of fishes.

⇒ Importance of Flora and Fauna:

- 1) We get crops which are used as food source.
- 2) Medicinal plants are obtained from flora.
- 3) Milching animals, which produce milk is used for different purposes.
- 4) Animals are used for meat, eggs and wool.
- 5) Animals are used for transportation purpose.

- 6) Fish is used as a nutritious diet.
- 7) Insects help in pollination of crops, fruits and Flowers.

→ Threats to flora and Fauna:

- 1) Hunting and poaching by greedy hunters.
- 2) Pollution caused by industrial waste.
- 3) Acid Rain.
- 4) Introduction of Alien species.
(Which is introduced from outside).
- 5) Over-exploitation by humans.
- 6) Deforestation - large scale cutting of trees.
- 7) Land over change. (Forest land changed to agricultural land).

⇒ Protection of Flora and Fauna:

1) Establishment of Biosphere reserves~

- Biosphere reserve Protected area where flora and Fauna is protected in their natural habitat.

⇒ Total 18 biosphere reserve in India.

- Out of which, 10 → included in World's network of biosphere reserve.

1) The Sunderbans. (2) The Nanda Devi. (3) The Gulf of Nicobar.

4) The Nilgiri. (5) The Nokrek. (6) The Great Nicobar.

• (7) The Similipal (Odisha) (8) The Pachmarhi (9) The

Pachmarhi. (10) The Panna. (11) The Achanakmar Amarkantak.

2) Establishment of botanical gardens.

↓
Is a place where plants are protected in artificial environment.

→ Govt. provides financial and technical assistance to support botanical gardens.

3) Important Projects:

- Some important projects are established by govt.

of India like:- 1) Project Tiger.

2) Project Rhino

3) Project Mangul, to conserve wildlife.

4) National parks and Wildlife Sanctuaries

are established by govt. to conserve wild life and flora in their natural habitat.

~~107~~ →

107 → National Park

573 → Wildlife Sanctuary } India.